

# Smart Motor Health Monitoring & Fault Detection System

Multi-sensor fusion and on-device decision making for real-time motor diagnostics — ESP32 · MPU6050 · ACS712 · Piezoelectric shock sensing.

|                    |                  |                  |           |
|--------------------|------------------|------------------|-----------|
| HEALTH INDEX RANGE | RESPONSE LATENCY | CLOUD DEPENDENCY | UNIT COST |
| 0-100%             | <1 sec           | NONE             | \$15      |

OVERVIEW

## What this system does

Electric motor failure — from overheating, mechanical wear, or sudden shock — accounts for over 80% of unplanned industrial downtime. This project puts diagnosis directly on the motor: an ESP32 microcontroller fuses four sensor streams into a single live **Health Index**, classifies the fault type locally, and cuts power automatically before damage occurs — no server round-trip required.

### Core capabilities

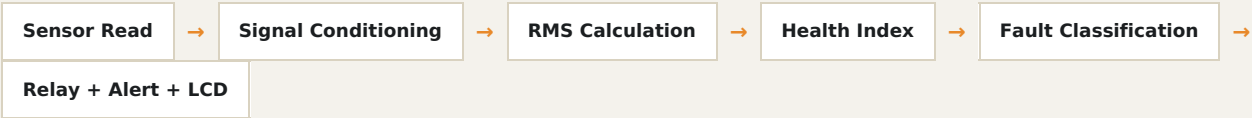
- Real-time fusion of temperature, current, and dual vibration sensing
- Fully on-device (edge) fault classification — zero cloud latency
- Automatic relay-based safety shutdown on severe faults
- Instant Telegram alerts over Wi-Fi
- Local 16×2 LCD status readout

### Why dual vibration sensing

The MPU6050 IMU catches broadband mechanical vibration (imbalance, misalignment). The piezoelectric disc catches sharp impulsive shocks the IMU misses. Together they discriminate fault types that single-sensor systems confuse.

ARCHITECTURE

## System processing pipeline



$$H = 100 \times (w_T \cdot S_T + w_I \cdot S_I + w_V \cdot S_V + w_P \cdot S_P)$$

WEIGHTED HEALTH INDEX ACROSS TEMPERATURE, CURRENT, MPU6050 & PIEZOELECTRIC CHANNELS

## Sensor & component stack

| Component               | Function                                  | Interface    |
|-------------------------|-------------------------------------------|--------------|
| ESP32 Dev Module        | Central processing, Wi-Fi, decision logic | —            |
| NTC Thermistor (10kΩ)   | Temperature / overheat detection          | Analogue ADC |
| ACS712-05B              | Current sensing / overload detection      | Analogue ADC |
| MPU6050                 | 6-axis vibration RMS (broadband)          | I²C          |
| Piezoelectric Disc      | Impulsive shock detection                 | Analogue ADC |
| 5V Relay Module         | Automatic motor power cutoff              | GPIO         |
| 16×2 LCD + I²C Backpack | Local status display                      | I²C          |

Validation

## Test results across fault conditions

| Condition        | Temp (°C) | Current (A) | MPU RMS | Piezo ADC | Health | Status   |
|------------------|-----------|-------------|---------|-----------|--------|----------|
| Normal           | 25        | 1.2         | 150     | 10        | 95%    | NORMAL   |
| Load             | 30        | 2.8         | 300     | 20        | 70%    | MILD     |
| Overload         | 34        | 5.0         | 600     | 38        | 40%    | ABNORMAL |
| Critical (MPU)   | 31        | 4.7         | 1200    | 30        | 10%    | SEVERE   |
| Critical (Piezo) | 27        | 3.8         | 350     | 900       | 15%    | SEVERE   |

Economics

## Bill of materials

| Component                | Qty | Unit Cost (INR) | Total (INR)    |
|--------------------------|-----|-----------------|----------------|
| ESP32 Dev Module         | 1   | 350             | 350            |
| ACS712-05B Module        | 1   | 80              | 80             |
| MPU6050 Module           | 1   | 70              | 70             |
| NTC Thermistor 10k       | 1   | 10              | 10             |
| Piezoelectric Disc       | 1   | 20              | 20             |
| 5V Relay Module          | 1   | 40              | 40             |
| 16×2 LCD + I²C           | 1   | 120             | 120            |
| 18650 Li-ion Cells       | 2   | 150             | 300            |
| DC Gear Motor            | 1   | 120             | 120            |
| Breadboard + Wires       | 1   | 100             | 100            |
| Resistors / Misc.        | —   | —               | 50             |
| Total Cost (Single Unit) |     |                 | ₹1,260 (~\$15) |

## Advantages over existing approaches

|                                                                                                                              |                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Edge Intelligence</b><br>All fault decisions run locally on the ESP32 — no cloud latency, sub-second protective response. | <b>Dual Vibration Sensing</b><br>MPU6050 + piezoelectric combination discriminates fault types single-modality systems miss. |
| <b>Low Cost</b><br>Complete prototype under \$15 — accessible for small-scale industrial and academic deployment.            | <b>Automatic Safety Shutdown</b><br>Relay-based hardware interlock is fail-safe, independent of software state.              |

ROADMAP

## Limitations & future scope

### Current limitations

- ACS712 zero-current offset drifts with temperature
- Health index weights are empirically assigned, not ML-optimized
- Validated on a single small DC motor only
- Requires Wi-Fi; no GSM fallback yet

### Planned enhancements

- On-device TinyML anomaly detection
- Custom PCB with EMI shielding
- MQTT / Grafana fleet dashboard
- Three-phase motor support (MCSA)